Cauchy–Binet gives a unique first nonzero term

rank r	first indices	contact s_r	coefficient
1	{1}	1	1
2	{1, 2}	3	1/8
3	{1, 2, 3}	6	1/1944
4	{1, 2, 3, 4}	10	1/31850496
5	{1, 2, 3, 4, 5}	15	1/62208000000000

$$s_r = 1 + 2 + \cdots + r = r(r+1)/2$$